A PROJECT REPORT ON FOOD QUALITY PREDICTION SYSTEM

Submitted in partial fulfilment of the requirement for the award of

BSC DEGREE IN ELECTRONICS OF KANNUR UNIVERSITY
BY

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2022 -2025

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2022-2024

CERTIFICATE

This is to certify that the project entitled " **FOOD QUALITY PREDICTION SYSTEM"** is a Bonafied record of the work done by MARWAN FAISAL AV,UJWAL K NAMBIAR,ABHINAV TV,AMRUTH V Students of Taliparamba Arts & Science College, Kanhirangad in partial fulfilment of the requirements for the award of the B.Sc.Degree in Electronics of Kannur University

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2.	2.

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ACKNOWLEDGEMENT

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For all this we are indebted to god, without his grace and benevolence we could not do nothing.

Place: KANHIRANGAD

Date:

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FOOD QUALITY PREDICTION SYSTEM

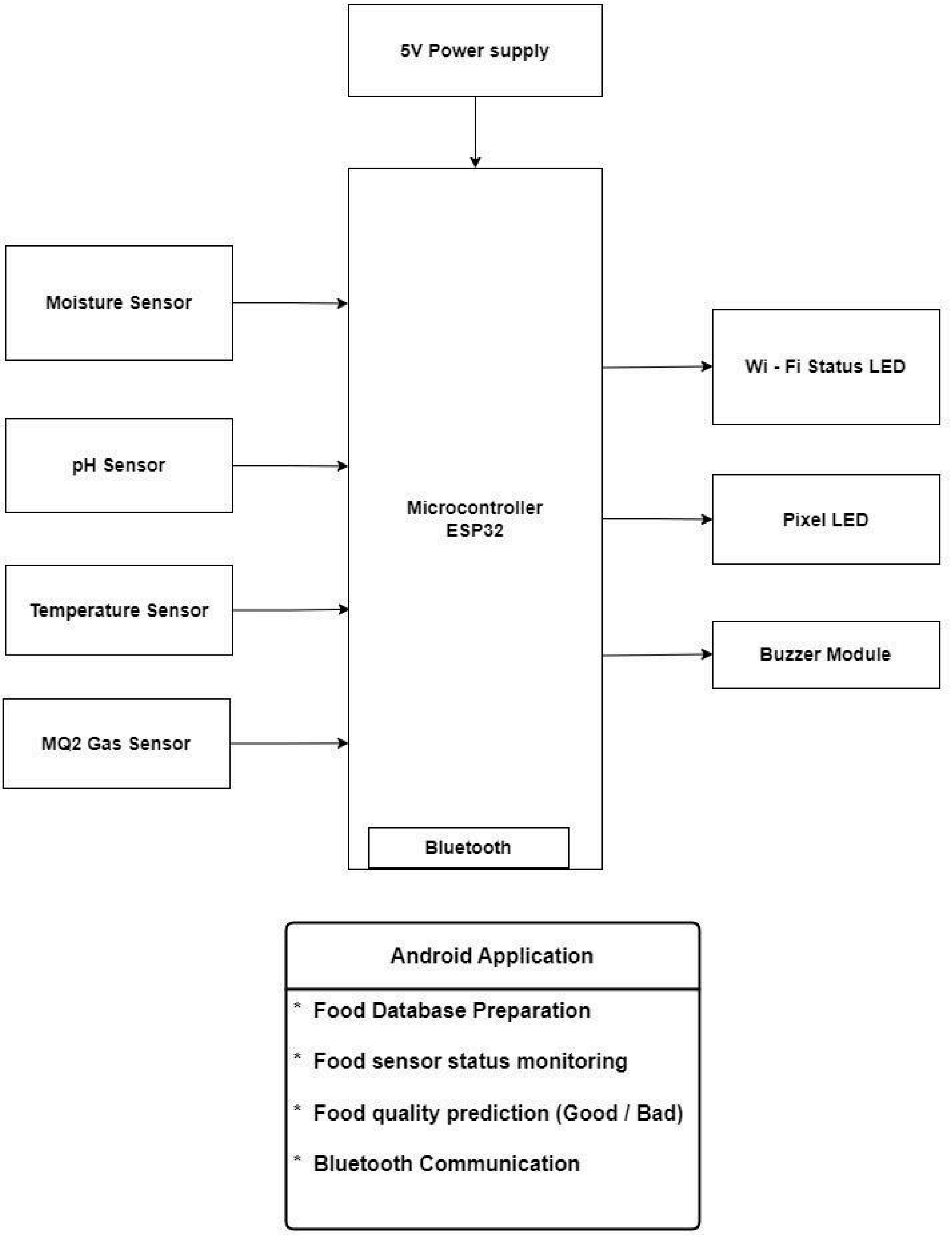
ABSTRACT

Food quality plays a crucial role in ensuring public health, and with increasing globalization, the need for efficient consumer products to monitor food safety has become essential. Recent trends highlight the rising consumption of altered, preserved, and contaminated food, which often leads to food poisoning and other health concerns. This paper proposes a food quality prediction system that utilizes multiple sensors to determine the state and quality of food accurately.

The system integrates an **ESP32 microcontroller** with a **moisture sensor**, **pH sensor**, **temperature sensor**, and **gas sensor** to monitor key parameters such as moisture levels, pH value, temperature, and the presence of harmful gases. Sensor data is transmitted to **Google Cloud** for real-time storage and validation against a predefined dataset to assess food quality. If contamination is detected, the system triggers a **warning sound** to alert users.

The proposed system is cost-effective, easy to implement, and suitable for common households, offering real-time monitoring and early detection of spoiled or contaminated food. This approach enhances food safety, helping users make informed decisions and minimize health risks.

BLOCK DIAGRAM



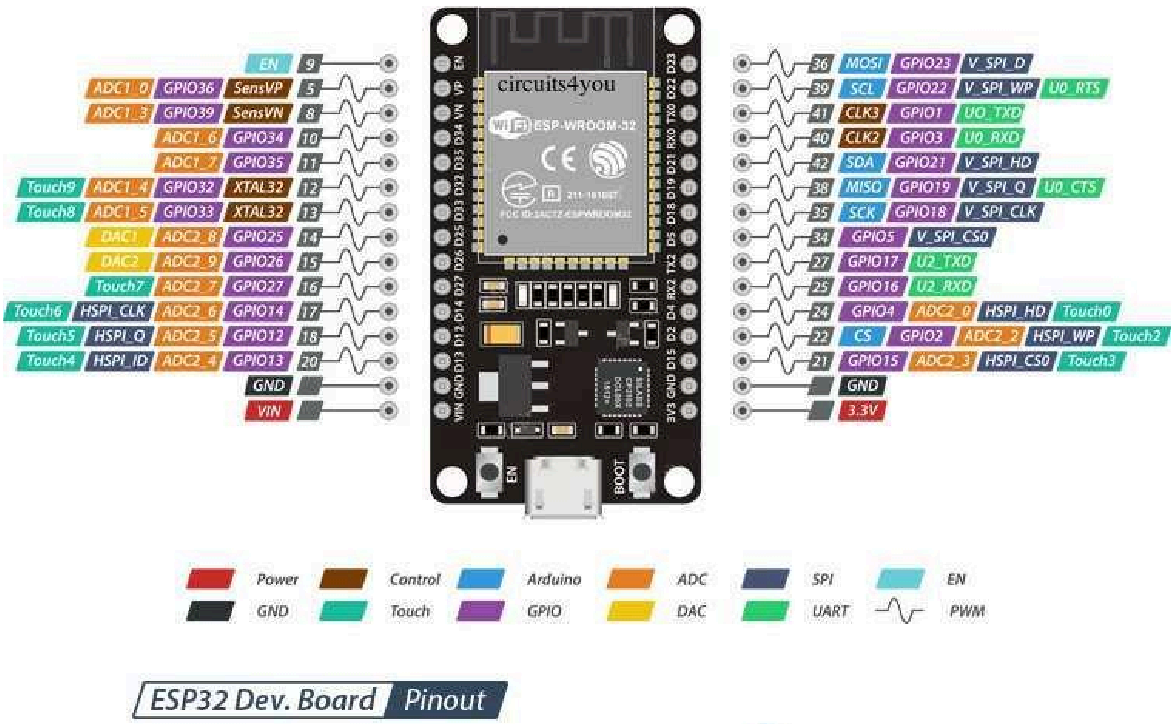
<u>**BLOCK DIAGRAM EXPLANATION**</u>

5V REGULATOR 7805:

The 7805 voltage regulator is a popular and reliable component used to provide a stable 5V DC output from a higher input voltage, typically ranging from 7V to 35V. It belongs to the 78xx series of fixed voltage regulators and is widely used in electronic circuits requiring a consistent 5V supply. The 7805 can deliver up to 1A of current, making it ideal for powering microcontrollers, sensors, and other digital devices. It also features built-in protection mechanisms, such as thermal shutdown and short-circuit protection, ensuring safe and stable operation even under varying load conditions. This regulator is commonly found in power supply circuits, providing essential voltage regulation for 5V electronics.

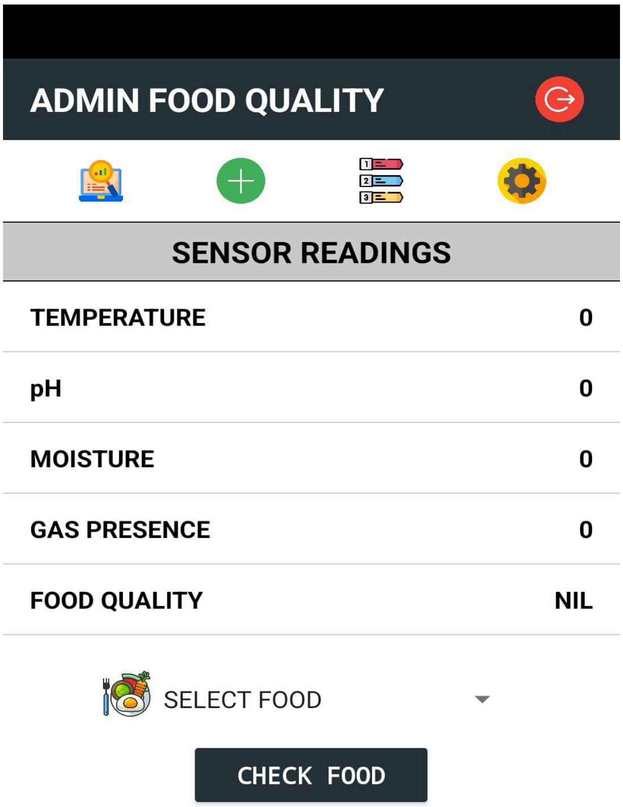
MICROCONTROLLER ESP32:

The ESP32 microcontroller is a highly versatile and powerful component extensively used in embedded systems and Internet of Things (IoT) applications. Developed by Espressif Systems, the ESP32 features a dual-core processor that operates at speeds up to 240 MHz, providing robust processing capabilities for multitasking and complex computations. Its integrated Wi-Fi and Bluetooth connectivity facilitate seamless communication with other devices and networks, enhancing its functionality in IoT projects. Additionally, the ESP32 offers a variety of flexible I/O options, including multiple GPIO pins, ADC channels, and communication interfaces like UART, SPI, and I2C, making it easy to integrate with various sensors and peripherals. Designed with energy efficiency in mind, it is ideal for battery-operated devices. The ESP32's combination of power, connectivity, and extensive community support has made it a popular choice among hobbyists and professionals for applications ranging from home automation to smart wearable technology.



MOBILE APPLICATION (MIT APP INVENTOR):

MIT App Inventor is a user-friendly platform developed by MIT that enables individuals with little to no programming experience to create Android applications through a visual, drag-and-drop interface. It was designed to democratize app development, making it accessible to students, educators, and hobbyists. Instead of writing traditional code, users can design the user interface and program the app's behaviour using building blocks that represent different functions. App Inventor supports real-time testing on connected devices, allowing developers to see immediate results. This platform is widely used for educational purposes, fostering creativity and problem-solving in a fun and intuitive way.



FIREBASE:

Firebase is a comprehensive app development platform developed by Google, designed to simplify the process of building and managing mobile and web applications. It provides a wide range of services and infrastructure, allowing developers to focus on creating a seamless user experience without worrying about backend complexities.

Firebase offers tools for real-time database management, authentication, cloud storage, and analytics, making it ideal for developers looking to scale their applications efficiently. One of its key features is the Realtime Database, which stores and syncs data between users in real time. This means that any changes made in the database are immediately reflected across all clients connected to the application. Firebase also offers Cloud Fire store, a scalable, flexible NoSQL database that supports richer queries and better performance for larger datasets.

In terms of security, Firebase includes user authentication services, allowing developers to authenticate users via email, phone, or third-party providers like Google, Facebook, and Twitter, all with minimal setup. Firebase also includes Firebase Hosting, which offers fast and secure hosting for web applications, as well as Cloud Functions, which allows developers to write server-side code triggered by Firebase features or HTTP requests.

Firebase's integration with Google Cloud Platform also enables smooth scaling and access to advanced cloud features, making it suitable for applications ranging from small-scale projects to enterprise-level systems.

TEMPERATURE SENSOR (DS18B20):

The DS18B20 is a digital temperature sensor known for its accuracy, reliability, and ease of use in various applications. Operating on a 1-Wire communication protocol, it requires only one data pin for communication with a microcontroller, making it highly efficient for IoT and embedded systems. The sensor can measure temperatures in the range of -55°C to +125°C with a typical accuracy of ±0.5°C and a resolution of 9-bit to 12-bit, which can be configured depending on the application. Its small size, low cost, and ability to operate in normal power mode or parasite power mode (using the data line for power) make it suitable for energy-efficient designs. Each DS18B20 sensor comes with a unique 64-bit serial code, allowing multiple sensors to connect to the same bus, which is ideal for applications like weather stations, industrial temperature monitoring, and home automation systems. Its versatility and precision make it a popular choice for temperature sensing in both simple DIY projects and complex industrial solutions.

MQ2 GAS SENSOR:

The MQ2 gas sensor is a versatile and widely used electronic sensor designed for detecting various gases such as LPG, propane, methane, alcohol, hydrogen, and smoke in the air. It operates on the principle of changes in resistance of its internal sensing material, typically tin dioxide (SnO_2), which has low conductivity in clean air but increases its conductivity when it comes into contact with gases. The sensor provides an analog output, which can be read by microcontrollers like Arduino and Raspberry Pi to measure gas concentration levels. The MQ2 sensor has a fast response time, high sensitivity, and low cost, making it ideal for applications such as gas leakage detection, fire alarm systems, and home safety devices. It requires a preheating period to stabilize for accurate readings and typically operates at 5V with a simple voltage divider circuit for signal processing. Its compact size, ease of use, and ability to detect multiple gases make it a popular choice in DIY electronics, IoT projects, and industrial safety solutions.

pH SENSOR:

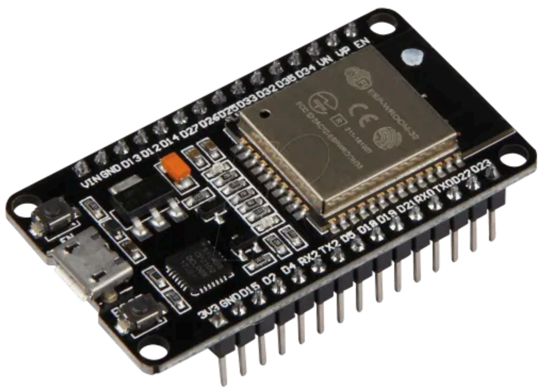
The pH sensor module, commonly used in electronics and embedded systems projects to measure the pH level of a solution, indicating its acidity or alkalinity. A pH sensor typically consists of two parts: the pH probe and the signal conditioning circuit board.

1. **pH Probe**: This is the part immersed in the liquid whose pH is to be measured. The probe generates a small voltage proportional to the pH of the solution, with acidic solutions generating a positive voltage and alkaline solutions generating a negative voltage. The probe often uses a glass bulb that is sensitive to hydrogen ion activity.

2**. Signal Conditioning Board**: Since the pH probe's voltage output is very low, it must be amplified for accurate reading. The signal conditioning board amplifies the voltage signal and outputs it in a form that can be read by a microcontroller, such as an Arduino or Raspberry Pi. The board also helps with calibration, allowing users to adjust the readings to match known pH levels using potentiometers on the circuit.

BLUETOOTH MODULE:

The Bluetooth module integrated into the ESP32 microcontroller provides robust wireless communication capabilities, supporting both Bluetooth Classic and Bluetooth Low Energy (BLE) protocols. This versatility makes the ESP32 ideal for a wide range of applications, from simple data transmission to complex Internet of Things (IoT) solutions. The dual-mode support allows for efficient communication with various Bluetooth-enabled devices, including smartphones, tablets, and sensors, enabling real-time data exchange. Additionally, the BLE functionality is particularly advantageous for battery-operated devices, as it offers low power consumption while maintaining connectivity. The ESP32's Bluetooth stack is well supported by various development frameworks, such as the Arduino IDE and ESP-IDF, which simplifies the development process for engineers and hobbyists. As a result, the ESP32's Bluetooth capabilities have made it a popular choice for applications in home automation, wearable technology, and health monitoring systems, facilitating the development of innovative wireless solutions.



5V DC SUPPLY:

A 5V DC supply provides a stable and direct current at a voltage level of 5volts, making it widely used in various applications, including automotive systems, consumer electronics, and renewable energy solutions. Unlike alternating current (AC), which alternates direction, direct current (DC) flows consistently in one direction, ensuring reliable power delivery for low-voltage devices. Common uses of 5V DC include powering car batteries, LED lighting, routers, and small appliances. Its steady output is essential for powering sensitive electronic circuits and is often used in systems where uninterrupted power is critical, such as solar energy storage and off-grid installations.

POWER LED:

A Power LED is a vital component in electronic devices, providing a visual signal that indicates whether the device is powered on and operational. Typically placed on the front panel or near the power switch, this low-power LED draws minimal current while effectively communicating the device's power status to the user. Its easy integration into various electronic systems, from household appliances to industrial equipment, makes it a common feature in many applications. Additionally, Power Indication LEDs are designed for long-lasting performance, ensuring they can provide continuous status updates without frequent replacements. By offering immediate visual confirmation of a device's power state, these LEDs facilitate user monitoring and troubleshooting, enhancing the overall user experience.

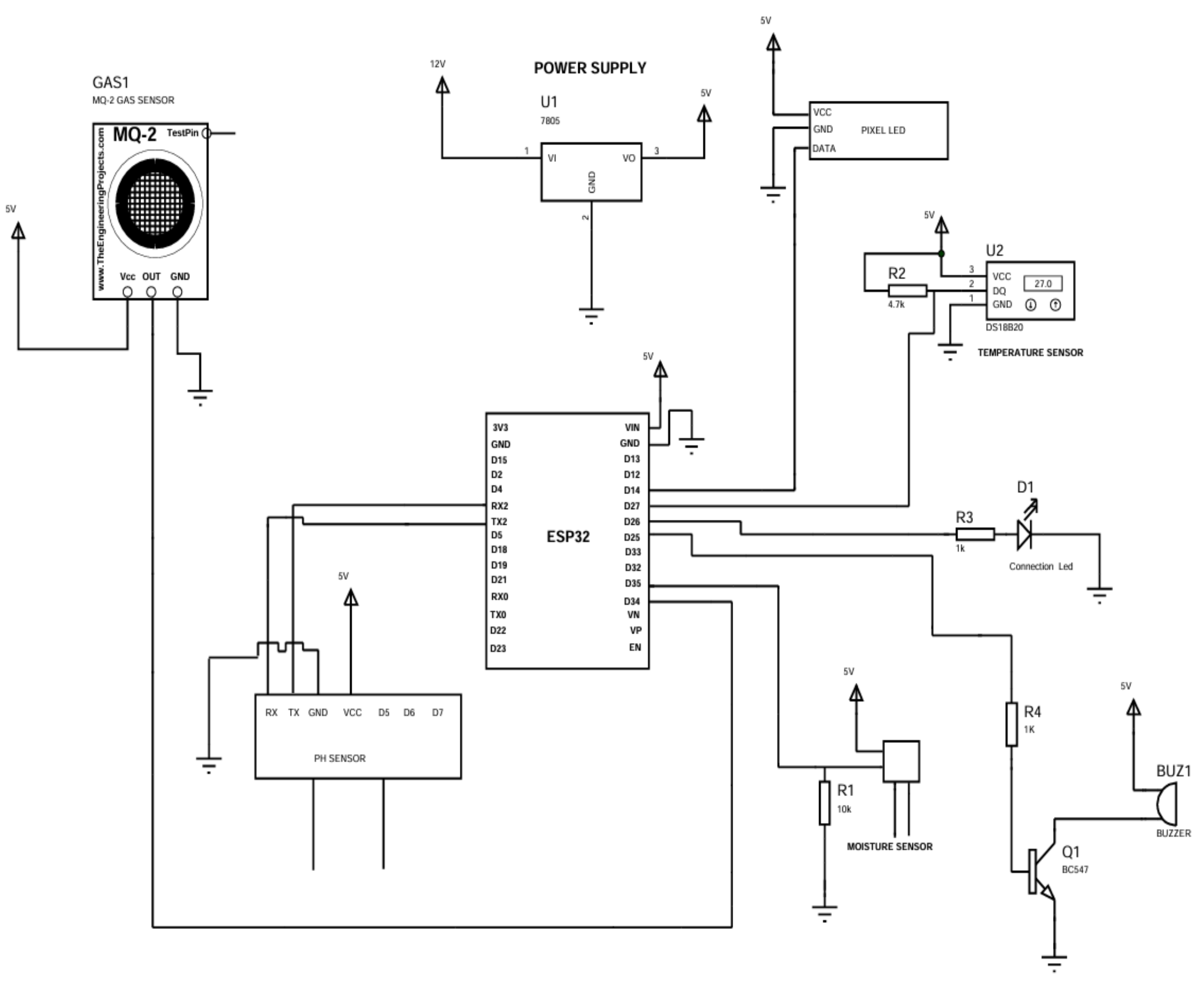
BUZZER:

A buzzer module is a compact electronic component used to produce sound in response to electrical signals. It is commonly used in a wide range of applications such as alarms, notifications, or audio indicators in projects and devices. The module typically consists of a piezoelectric buzzer or electromagnetic buzzer that can generate different tones or beeps when voltage is applied. It can be controlled easily with a microcontroller like an Arduino, using either digital signals for simple on/off beeps or PWM (Pulse Width Modulation) for producing varying tones. Buzzer modules are widely used in alarm systems, timers, and interactive electronics due to their simplicity, low cost, and effectiveness in providing auditory feedback.

NEO PIXEL LED:

Neo Pixel LEDs are individually addressable RGB LEDs housed in a strip or matrix, controlled by a single data line. Manufactured by Adafruit, these LEDs allow precise control over the colour and brightness of each LED, enabling a wide array of lighting effects, animations, and patterns. Neo Pixels are built using WS2812 or SK6812 LED chips, which integrate both the LED and the control circuitry. They operate on a 5V power supply and communicate using a serial protocol, making them easy to use with microcontrollers like Arduino or Raspberry Pi. With vibrant colours, simple wiring, and the ability to control hundreds of LEDs with just one pin, Neo Pixels are widely used in decorative lighting, displays, and interactive projects.

CIRCUIT DIAGRAM



WORKING

The **Food Quality Prediction System** is designed to ensure food safety by continuously monitoring key parameters that indicate spoilage or contamination. The system integrates an **ESP32 microcontroller** with multiple sensors, including a **moisture sensor, pH sensor, temperature sensor, and gas sensor**, to assess the freshness of food items. These sensors collect real-time data on moisture levels, acidity, temperature, and the presence of harmful gases, which are then processed by the microcontroller. The collected data is transmitted to **Google Cloud**, where it is validated against a predefined dataset to determine food quality. If contamination is detected based on abnormal sensor readings, the system triggers a **warning sound** to alert users about potential spoilage. Additionally, users can monitor food quality remotely through a **mobile application or web interface**, receiving instant notifications about any detected issues. This cost-effective and user-friendly solution is ideal for households, restaurants, and food industries, helping to reduce foodborne illnesses and prevent wastage. By offering **real-time monitoring, cloud-based analysis, and early detection of spoilage**, the system enhances food safety and empowers consumers to make informed decisions about their food consumption.

ALGORITHM

Step 1: Start

Step 2: Initialize all pins.

Step 3: Read moisture sensor value from moisture sensor.

Step 4: Read gas sensor value from gas sensor.

Step 5: Read temperature value from temperature sensor.

Step 6: Read pH sensor value from pH sensor.

Step 7: Send sensor data through serially.

Step 8: Check if character is available through serially, then go to step 9. Else go

to step 3.

Step 9: If character received is ‘G’, then turn on green colour in pixel LED. Else

go to step 10.

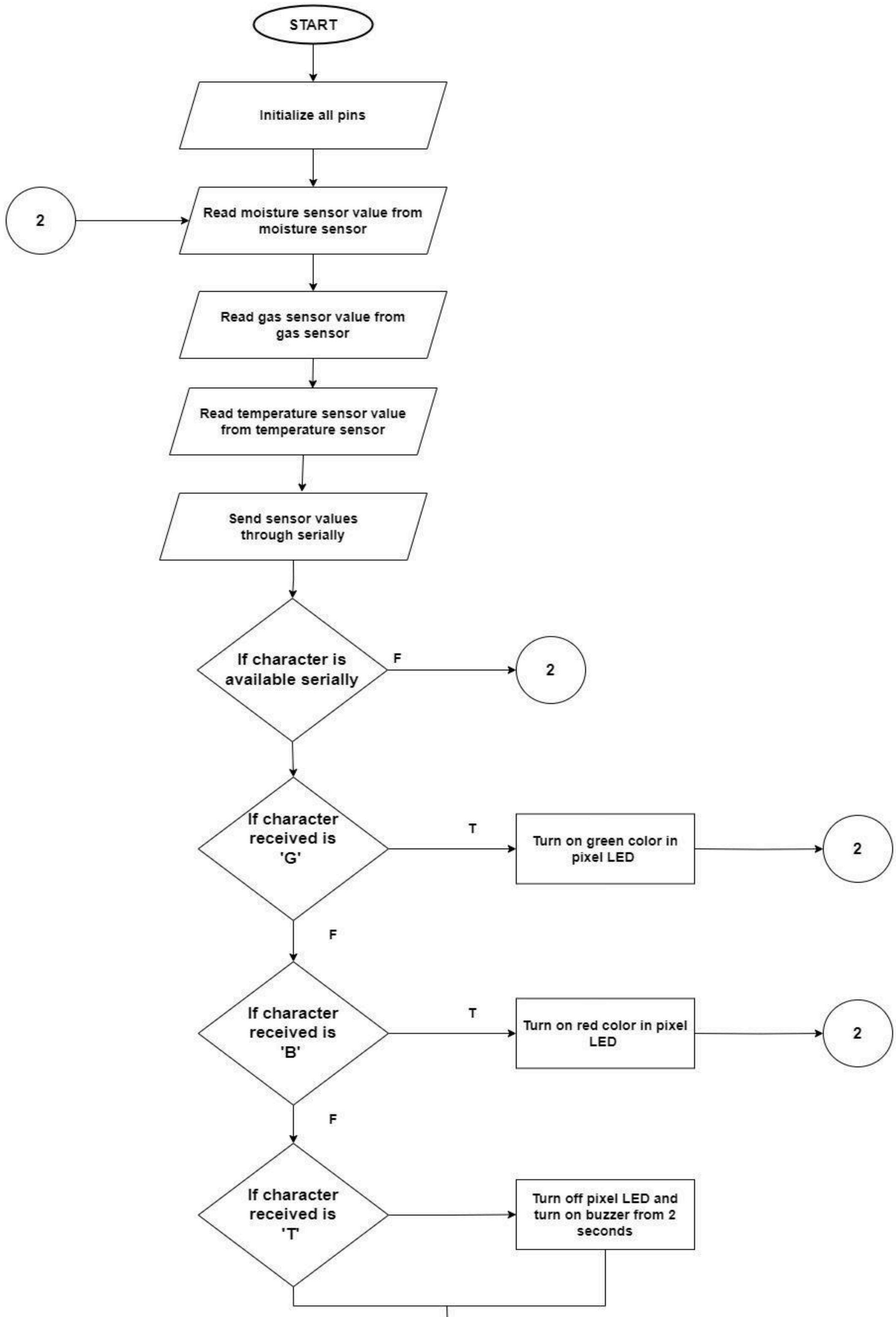
Step 10: If character received is ‘B’, then turn on red colour in pixel LED.

Else go to step 11.

Step 11: If character received is ‘T’, then turn off pixel LED and turn on buzzer

for 2 seconds. Else go to step 3.

FLOWCHART


START
Initialize all pins
2
Read moisture sensor value from moisture sensor
Read gas sensor value from gas sensor
Read temperature sensor value from temperature sensor
Send sensor values through serially
If character is available serially
F
2
If character received is 'G'
T
Turn on green color in pixel LED
2
F
If character received is 'B'
T
Turn on red color in pixel LED
2
F
If character received is 'T'
Turn off pixel LED and turn on buzzer from 2 seconds

ADVANTAGES

- **Real-Time Monitoring**
  - Continuously tracks food quality parameters such as moisture, pH, temperature, and gas levels.
  - Provides instant updates to users for timely action.
- **Early Spoilage Detection**
  - Identifies signs of food contamination before it becomes harmful.
  - Helps prevent foodborne illnesses and ensures consumer safety.
- **Cloud-Based Analysis**
  - Sensor data is stored and validated in **Google Cloud**, allowing for accurate quality assessment.
  - Enables remote access to food quality reports via a mobile app or web dashboard.
- **User-Friendly Alerts**
  - Triggers a **warning sound** when food contamination is detected.
  - Sends **notifications** to users, enabling quick decision-making.
- **Cost-Effective Solution**
  - Uses affordable and widely available sensors, making it accessible for households and food industries.
  - Reduces food wastage by detecting spoilage early.
- **Enhanced Food Safety**

- Helps maintain proper storage conditions by monitoring temperature and moisture.

LIMITATIONS

Limited Sensor Accuracy

- The accuracy of **moisture, pH, temperature, and gas sensors** may vary due to environmental factors.
- Calibration and periodic maintenance are required for reliable results.

- **Dependency on Predefined Data**
  - The system relies on a **predefined dataset** for food quality validation.
  - It may not accurately detect **new types of food spoilage** not included in the dataset.
- **Potential False Alarms**
  - Sudden temperature changes or external gases may trigger **false contamination alerts**.
  - Users might need to manually verify food spoilage in some cases.
- **Limited Detection Scope**
  - The system can detect basic spoilage indicators but **may not identify specific bacteria or toxins** like Salmonella or E. coli.
  - Laboratory testing would still be required for detailed microbial analysis.
- **Cloud Dependency**

- Requires an **active internet connection** for real-time data transmission and cloud analysis.
- If the internet is unavailable, data processing may be delayed.

FUTURE APPLICATION

- **Smart Refrigerators**
  - Integration with refrigerators to monitor stored food quality and provide expiration alerts.
  - Automated temperature and humidity adjustments to prolong food freshness.
- **Food Supply Chain Monitoring**
  - Implementation in **food transportation and storage** to ensure quality control during transit.
  - Reducing food wastage in supply chains by detecting spoilage at early stages.
- **Supermarket and Grocery Stores**
  - Used in supermarkets to monitor the freshness of perishable products like dairy, meat, and vegetables.
  - Alerts store managers about spoiled or expired products to maintain food safety standards.
- **Food Processing and Packaging Industries**
  - Ensuring food safety in **processing plants and packaging units** by monitoring environmental conditions.

- Detecting contamination during manufacturing and improving **quality control processes**.

CONCLUSION

The **Food Quality Prediction System** is a smart, efficient, and cost-effective solution for ensuring food safety by continuously monitoring key parameters such as **moisture, pH, temperature, and gas levels**. By integrating an **ESP32 microcontroller** with multiple sensors and **Google Cloud storage**, the system provides real-time analysis, helping users detect spoilage and contamination at an early stage. Its ability to generate **alerts and notifications** enables consumers to make informed decisions, reducing the risk of foodborne illnesses and minimizing food waste.

Despite some limitations, such as **sensor accuracy, cloud dependency, and detection scope**, the system remains highly valuable for applications in **households, food industries, supermarkets, and healthcare sectors**. Future advancements, including **AI integration, IoT connectivity, and supply chain monitoring**, can further enhance its capabilities.

In conclusion, the **Food Quality Prediction System** represents a significant step toward **improving food safety standards**, offering a practical and innovative approach to preventing contamination and ensuring healthier food consumption for individuals and businesses alike.

PROGRAM CODE

```
#include "BluetoothSerial.h"
BluetoothSerial SerialBT;

#include <Adafruit_NeoPixel.h>
#define PIN 27
#define NUMPIXELS 5
Adafruit_NeoPixel pixels(NUMPIXELS, PIN, NEO_GRB + NEO_KHZ800);

#include <OneWire.h>
#include <DallasTemperature.h>
#define ONE_WIRE_BUS 14
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature sensors(&oneWire);
int numberOfDevices;
DeviceAddress tempDeviceAddress;

HardwareSerial mySerial(2);
```

```
int moisture_sensor_pin = 32, Gas_sensor_pin = 35, Buzzer = 25;

int moisture_sensor_status = 0, Gas_sensor_status = 0;

String data;
unsigned long ms, msLast;
float tempC = 0;

String ph_value_string, ldr_value_string, temp_value_string, ph_sensor_data;
float ph_data = 0, ldr_data = 0, temp_data = 0;

int connection_led = 26, connection_interval = 250;
unsigned long connection_millis = 0;

void setup()
{
  mySerial.begin(9600, SERIAL_8N1, 16, 17);

  pinMode(moisture_sensor_pin, INPUT);
  pinMode(Gas_sensor_pin, INPUT);
  pinMode(Buzzer, OUTPUT);
  pinMode(connection_led, OUTPUT);
```

```
  SerialBT.begin("FOOD-QUALITY");

  sensors.begin();

  numberOfDevices = sensors.getDeviceCount();

  pixels.begin();
  for (int i = 0; i < NUMPIXELS; i++)
  {
    pixels.setPixelColor(i, pixels.Color(255, 255, 255));
  }
  pixels.show();
}

void loop()
{
  if (SerialBT.hasClient())
  {
    digitalWrite(connection_led,!digitalRead(connection_led));
  }
  else
  {
    digitalWrite(connection_led,HIGH);
  }
```

```
moisture_sensor_status = analogRead(moisture_sensor_pin);
Gas_sensor_status = analogRead(Gas_sensor_pin);

sensors.requestTemperatures();
for (int i = 0; i < numberOfDevices; i++)
{
  if (sensors.getAddress(tempDeviceAddress, i))
  {
    tempC = sensors.getTempC(tempDeviceAddress);
  }
}

if (mySerial.available() > 0)
{
  delay(100);
  String c = mySerial.readString();

  ph_value_string = c.substring(5, 10);
  ldr_value_string = c.substring(21, 24);
  temp_value_string = c.substring(29, 32);

  ph_data = ph_value_string.toFloat();
  ldr_data = ldr_value_string.toFloat();
```

```
  temp_data = temp_value_string.toFloat();

  ph_sensor_data = "*P" + String(ph_data) + "*" + "*D" + String(ldr_data) +
"*" + "T" + String(temp_data) + "*";  }

 if (SerialBT.available() > 0)
 {
  char rec = SerialBT.read();
  if (rec == 'G')
  {
   for (int i = 0; i < NUMPIXELS; i++)
   {
    pixels.setPixelColor(i, pixels.Color(0, 255, 0));
   }
   pixels.show();
  }
  if (rec == 'B')
  {
   for (int i = 0; i < NUMPIXELS; i++)
   {
    pixels.setPixelColor(i, pixels.Color(255, 0, 0));
   }
   pixels.show();
  }
```

```
    if (rec == 'T')
    {
      for (int i = 0; i < NUMPIXELS; i++)
      {
        pixels.setPixelColor(i, pixels.Color(255, 255, 255));
      }
      pixels.show();

      digitalWrite(Buzzer, HIGH);
      delay(300);
      digitalWrite(Buzzer, LOW);
      delay(300);
      digitalWrite(Buzzer, HIGH);
      delay(300);
      digitalWrite(Buzzer, LOW);
      delay(300);

    }
  }

  data = "*M" + String(moisture_sensor_status) + "*G" +
String(Gas_sensor_status) + "*T" + String(tempC) + "*P" + String(ph_data) +
"*";

  ms = millis();
```

```
 if (ms - msLast > 1000)
 {
   SerialBT.println(data);
   msLast = ms;
}
```

REFERENCE

Website:

- WIKIPEDIA - https://en.wikipedia.org/wiki/Main_Page
- GOOGLE-http://www.google.com/